



CDS 6324
DATA VISUALIZATION

ASSIGNMENT (30%)
Lecture Section: TC1L
Tutorial Section: TT2L
Group Number: TT2L_G15
Submitted to: Dr. Arselan Ashraf

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1.0 Introduction

Malaysian travellers face a telecommunications market with many providers when planning international trips. Eight major brands: Yes, Tune Talk, U Mobile, redONE, Unifi Mobile, Maxis, Hotlink, and CelcomDigi, collectively offer over sixty travel-oriented plans, each with different country coverage, pricing structures, validity periods, and inclusions. Without a unified view, a traveller cannot easily answer the most fundamental question: *which plan is best for my trip?* Our project addresses this gap. We built a Power BI dashboard that combines roaming pass data, per-country IDD calling rates, and a country dimension table into a single interactive decision-support tool.

2.0 Data Collection

2.1 Data Sources

The project draws from public data, each cited and dated to ensure reproducibility :

1. **travelplans.csv** (65 rows × 17 columns) : The fact table containing plan attributes. Sourced from the eight Malaysian telco brand websites providing roaming and travel plans. Each plans' data are collected and verified manually into one shared excel file.
2. **countries.csv** (256 rows × 6 columns) : The country dimension with ISO codes, dialing codes, region, and local operators. Sourced from International reference standards which are *ISO 3166-1* (from the International Organization for Standardization) and *ITU-T E.164* (from the International Telecommunication Union)
3. **idd.csv** (2,980 rows × 11 columns) : Per-country international dialing rates by brand.

2.1.1 Scraping Methodology for IDD

The eight brand websites varied significantly in structure, thus requiring different extraction strategies. The process requires eight Python scraper scripts for collection. The finalised idd.csv comes from consolidating all eight source files, with post-processing fixes applied automatically.

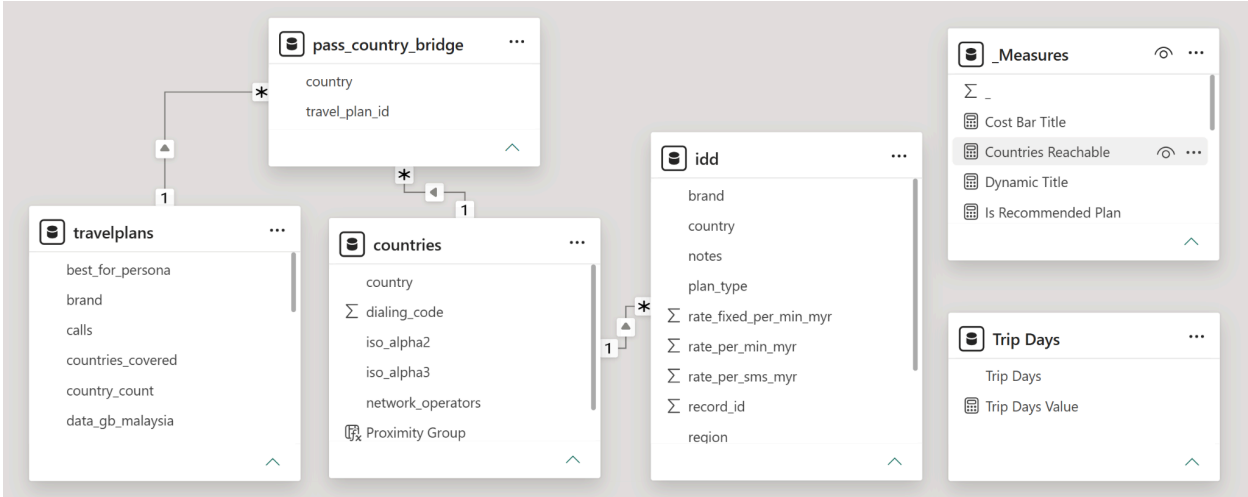
Brand	Strategy	Tool
Yes Mobile	Extracted JSON directly from HTML from embedded var jsonIdd, no browser needed	Python + regex parser
Maxis, Hotlink	Playwright headless browser, select country, click each tab, parse active pane only	Playwright + BeautifulSoup
Unifi	Playwright with scoped .dropdown-menu.show selector to avoid matching navigation links	Playwright
CelcomDigi	Playwright select + regex rate extraction from rate blocks	Playwright + regex

U Mobile, Tune Talk, redONE	Playwright dropdown selection, rate card parsing, custom sequential h2/h5 element parser	Playwright + custom parser
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2.2 Data Schema Design

The three source tables were connected in Power BI using a star schema, with *countries* as the central dimension and *travelplans* and *idd* as fact tables, both joined through the country name field. The *travelplans* table stores country coverage as a pipe-separated string per plan, which cannot be directly joined. We resolved this in Power Query by splitting the *countries_covered* column by the pipe delimiter into individual rows, creating a bridge table called *pass_country_bridge*.

All relationships were set to single-direction cross-filtering to prevent ambiguous filter paths. This means clicking a country on the map propagates the filter through the bridge to *travelplans* and separately to *idd*, updating every visual on the page in one interaction.



2.3 Data Preparation

There were preparation tasks in our Power Query audit trail :

- Brand-name standardisation:** *travelplans* used *unifi*, *umobile*, *redOne*, *TuneTalk*; *idd* used *Unifi*, *U Mobile*, *redONE*, *Tune Talk*. We normalised *travelplans* to match *idd*'s canonical spellings using Replace Values to ensure the tables could be linked without issue.
- Pass-country bridge construction:** we referenced *travelplans*, split *countries_covered* by the ' | ' delimiter into rows, and trimmed whitespace. This created the bridge table that enables country-selection on the Map visual and all other measures.
- Country dimension aliasing:** we added 18 alias rows to *countries.csv* (e.g., "Türkiye" → Turkey, "UAE" → United Arab Emirates) sharing the same ISO codes and operators as their canonical entries.

2.3.1 Derived Metrics

To support meaningful comparison and recommendation, we derived additional metrics that the raw source data alone could not provide.

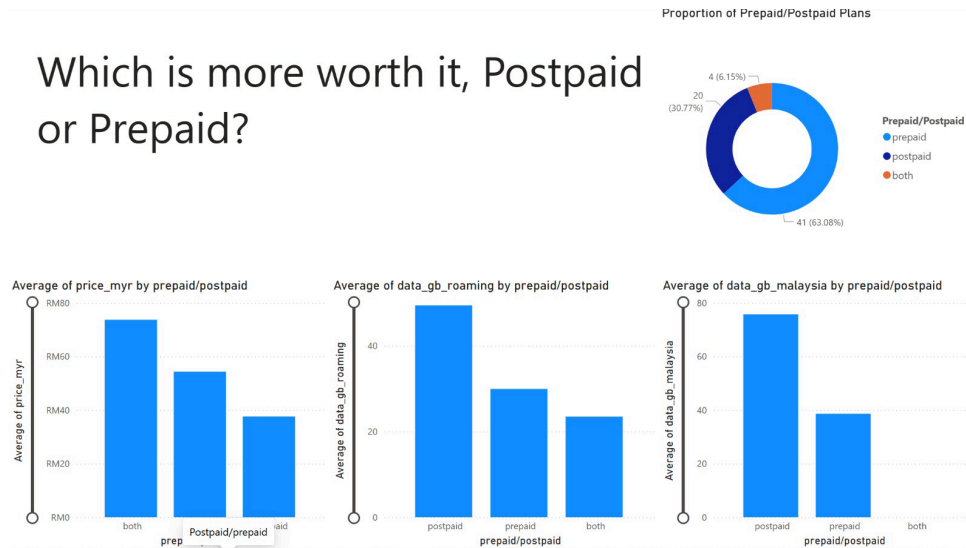
1. **Price per GB of Roaming (*price_per_gb_roaming*):** a normalised cost metric calculated as $\text{price_myr} / \text{data_gb_roaming}$.
2. **Price per Day (*price_per_day*):** calculated as $\text{price_myr} / \text{validity_days}$. Used in the duration-aware recommendation logic to support short-trip use cases.
3. **Travel Value Score (*composite, 0–100*):** a calculated column on travelplans that ranks each plan on overall travel utility beyond price alone. The score is a weighted composite of six normalised components: roaming data (30%), countries covered (25%), calls included (15%), validity (15%), hotspot (10%), and travel insurance (5%). Each component is capped at 1.0 before weighting to prevent high-outlier plans from distorting the scale. The weights reflect the hierarchy brands themselves use in their marketing, as data allowance and destination breadth are consistently headlined, while insurance and hotspot appear as secondary features.
4. **Effective Call Rate (*effective_call_rate*):** a DAX measure that computes the real cost of calling from a destination by returning zero when a plan includes calls, and falling back to the brand's published IDD per-minute rate from the idd table when calls are not included. This bridges the two fact tables into a single decision-relevant figure, surfacing the hidden cost that plan pricing alone does not capture.
5. **Trip Days Parameter (*Trip Days Value*):** a What-If parameter table generated in Power BI's Modeling pane, producing a numeric range from 1 to 50 representing the user's intended trip duration in days. The parameter feeds a DAX measure that filters the eligible plan pool to only plans where *validity_days* is greater than or equal to the selected value, then applies a nearest-ceiling logic to prioritise plans closest to the trip duration rather than overshooting to the longest available plan. This transforms a static dataset into a duration-aware recommendation engine without requiring any changes to the underlying data.

3.0 Exploratory Data Analysis (EDA)

3.1 Question 1 : Which is more worth it, Postpaid or Prepaid?

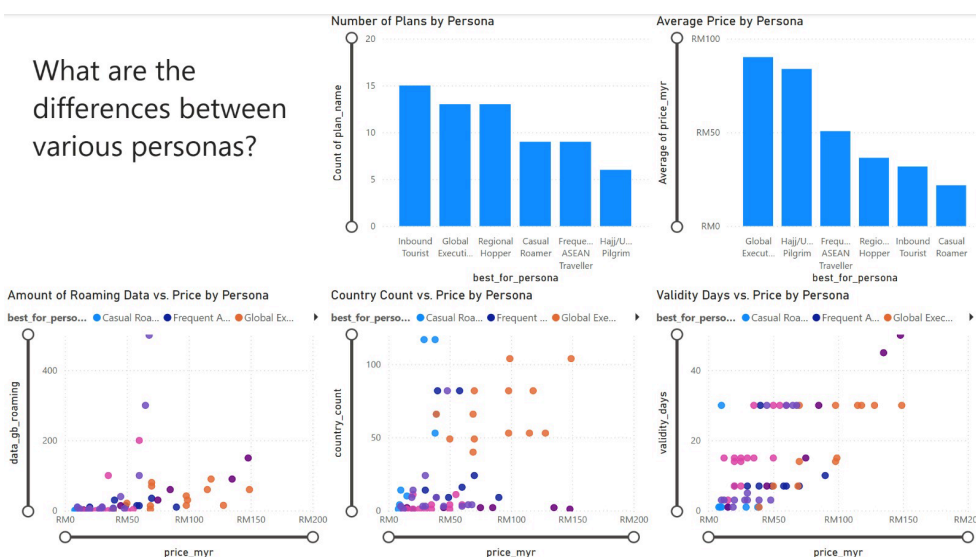
Postpaid plans offer the strongest value on paper. Prepaid plans sit in the middle on all three dimensions. Plans marketed as both prepaid and postpaid are the most expensive on average despite offering less data than postpaid, likely reflecting their flexibility premium. However, postpaid plans in Malaysia typically require a monthly contract commitment, which reduces flexibility for one-off or infrequent travellers. For this reason, the dashboard does not filter by plan type by default as the right choice depends on whether the user already holds a postpaid subscription or is purchasing specifically for a single trip.

Which is more worth it, Postpaid or Prepaid?



3.2 Question 2 : What are the differences between various personas?

In terms of number of plans by persona, the number varied somewhat between personas with the most plans belonging to the Inbound Tourist persona, and the least belonging to the Hajj/Umrah Pilgrim persona. As expected, the Global Executive persona had the most expensive plans on average and Casual Roamers had the least expensive plans on average. In terms of roaming data vs price, it can be roughly extrapolated that there is correlation between amount of roaming data and price. There doesn't seem to be a lot of correlation between the price and the number of countries covered, although the most number of countries covered belonged to the Casual Roamers and Global Executives. Furthermore, when it comes to Validity Days, there is some level of correlation between the price and validity days, but generally plans end after common periods of 1, 2, 3, 7, 14 or 30 days with some outliers.



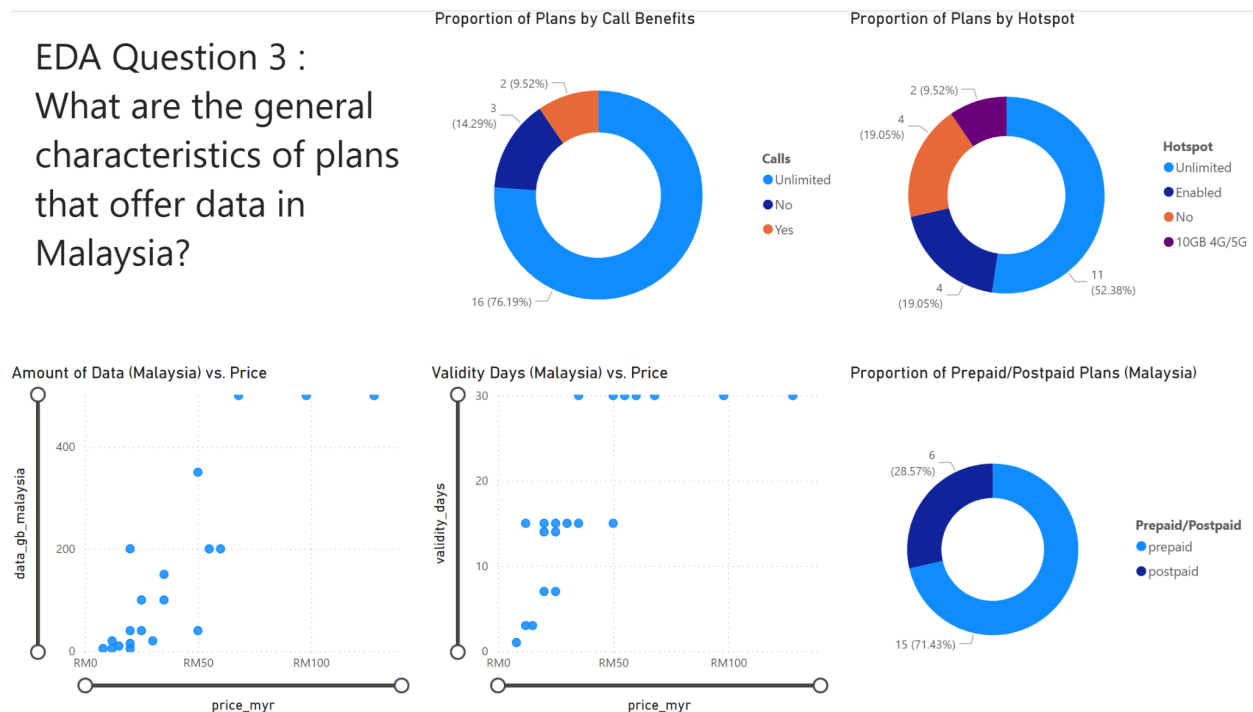
3.3 Question 3 : What are the general characteristics of plans that offer data in Malaysia?

In this analysis, we attempted to find the general characteristics of plans designed for inbound tourists to Malaysia. This analysis shouldn't be confused with an analysis focussed on plans for residents of Malaysia, as most of the plans in our data are targeting tourists/frequent travellers.

Most plans (76%) offered unlimited (domestic) calls, those falling under the "Yes" category indicates that these plans offered some level of benefits, but not unlimited. For mobile hotspots, half of the plans offered unlimited (no restrictions to hotspot, limited however by the plan's data quota); while 4 plans had hotspots "Enabled" means some level of restrictions were applied; another 2 plans allocated only 10GB for hotspots; and lastly 4 plans did not permit mobile hotspotting. Roughly 71% (15) of all plans which offered data in Malaysia were prepaid, with only 6 plans being postpaid; this approximately mirrors the proportion of plans by prepaid/postpaid shown in the previous analysis.

When it came to amount of data vs price, although there are a few 500GB outliers, most plans offered less than 100 GB of data and cost below RM 30. With the plans following a roughly linear pattern indicating a positive correlation between amount of data and price. In terms of validity days against price, plans had a range of between 1 to 30 days, and again a roughly linear pattern emerged, indicating a positive correlation between validity days and price.

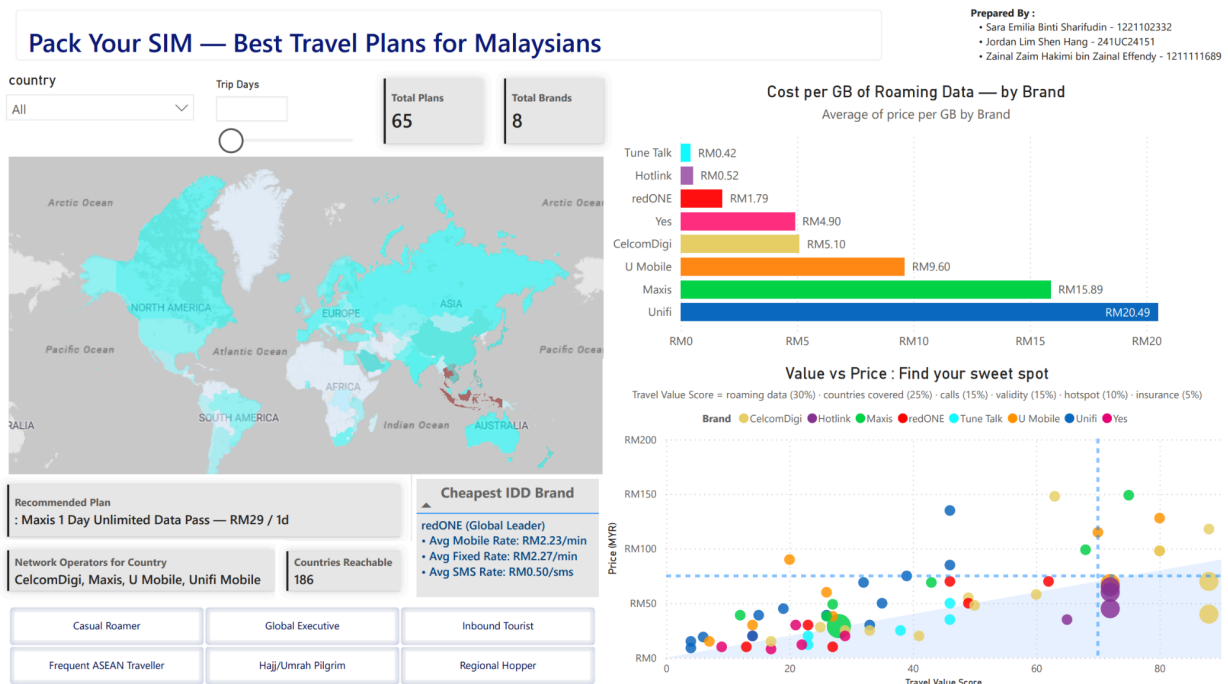
EDA Question 3 :
What are the general characteristics of plans that offer data in Malaysia?



4.0 Dashboard Design

4.1 Main Layout Walkthrough

The dashboard structure follows Cole Nussbaumer Knaflic's three-part narrative arc from Storytelling with Data, where every visualisation serves a specific role in guiding the user toward a decision rather than presenting raw data for the user to interpret independently.



The dashboard is built on a single page with five visual zones following below :

Visual Zone	Element	Narrative Role	Purpose
Header strip	Dynamic title, Trip Days slider, country dropdown, 2 KPI cards	Beginning (Attract)	Opens with Total Plans and Total Brands, giving the user a sense of the market before any interaction. The country dropdown and Trip Days slider signal that the user controls the analysis.
Driver map	Filled choropleth, teal to purple gradient by plan count	Beginning (Attract)	The dominant visual on the page. Anchors the narrative around destination first because every travel decision starts with where, not how much.
Right column	Cost bar chart, Value vs Price	Middle (Engage)	Clicking a country re-ranks brands by price per GB on the bar chart and filters the scatter to plans

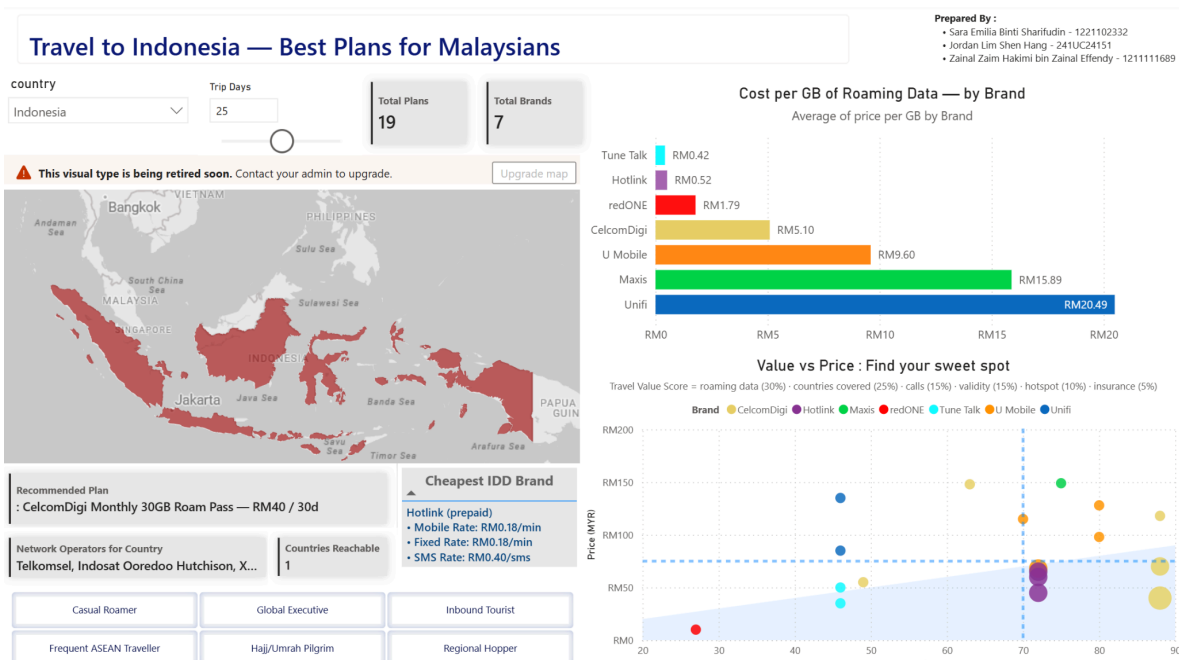
	scatter with sweet-spot quadrant		covering that destination. Dot size on the scatter indicates the recommended plan.
Persona buttons	Six tappable persona tiles	Middle (Engage)	Refines the recommendation by travel intent without requiring the user to know which specific plan to search for. Works independently of country selection.
Decision panel	Recommended Plan, Cheapest IDD Brand, Network Operators cards	End (Punchline)	Surfaces a single named plan with price and validity based on the selected country, persona, and trip duration. The Network Operators card completes the picture by showing which local networks the user will roam onto.

4.2 Dashboard Interactivity

Four cross-filter mechanisms are present on the dashboard:

1. Country selected → filter all visuals to that country's plans
2. Click on brand bar → filters, recommendation card, and IDD card switch to that brand
3. Click on a plan in the scatter plots → override the recommendation card to show "Selected Plan: ..." with the clicked plan's details
4. Persona button click → filters the eligible pool to plans matching that persona, ignoring trip days.

The Validity Days parameter slider (defaulting to 1, allowing all plans on load) filters the eligible pool by validity_days >= trip days, with nearest-ceiling logic so a 3-day trip recommends a 7-day plan rather than overshooting to a 30-day plan.



4.3 Data Storytelling

The issue faced by Malaysian travellers is how to navigate a market with 8 major telco brands offering countless plans and make a decision on the best plan for them. This overwhelming amount of plans makes it difficult and time consuming for travellers to make comparisons.

On first load, the dashboard gives an immediate overview of the entire market. The world map displays a heatmap showing which countries are covered by the most plans, the bar chart ranks all eight brands by average price per GB of roaming data, and the scatter plot positions all plans on a value versus price axis using the Travel Value Score. Plans that fall in the sweet spot quadrant, defined as a Travel Value Score of 70 or above and a price of RM75 or below, are rendered with larger dots to draw immediate attention to the plans that offer disproportionate value relative to their cost. Users can hover over any dot to see the plan name, brand, score, price, and key benefits.

When the user knows their destination, they select a country from the dropdown or click directly on the map. Every visual updates to show only plans covering that country, the bar chart re-ranks brands for that destination, and the Recommended Plan card surfaces the single best-value plan based on the selected country and trip duration entered through the Trip Days slider. The Cheapest IDD card and Network Operators card load alongside, giving the user the full cost picture including calling rates and which local networks they will connect to. For users who do not yet have a destination in mind but know their travel purpose, the six persona buttons provide an alternative entry point.

The story concludes with the actionable conclusion delivered by the Decision Panel. After the user refines their search using the Countries selection, Trip Days slider and persona buttons, the panel displays the recommendation in the Recommended Plan card. This delivers a suitable plan with supporting intelligence such as the cheapest IDD calling rate and the local network operators a traveller will roam onto, allowing a complete data-driven decision to be made in under a minute.

5.0 Conclusion

Our project addresses the issue of travellers being unable to effectively compare plans across a multitude of plans without significant effort. We built a Power BI dashboard that combines roaming pass data, per-country IDD calling rates, and a country dimension table into a single interactive decision support tool. The dashboard helps a Malaysian traveller whether a Hajj pilgrim, ASEAN business traveller, or casual roamer to identify the most cost-effective plan for their destination, trip duration, and travel style.

The dashboard's central interaction is a clickable world map. When a user selects a destination country, the dashboard recommends the best-value plan, surfaces the cheapest IDD calling rate from any brand for that destination, and displays the local network operators a traveller will roam onto. Persona buttons further refine the recommendation to match the user's travel pattern. The complete decision can be made in under a minute.

Group Contribution Declaration

Lecture Section: TC1L

Tutorial Section: TT2L

Group Number: TT2L_G15

Task Distribution

Student Name	Task Done	Contribution Percentage
Sara Emilia Binti Sharifudin	• Data collection and web scraping • DAX measures and cross-filter interaction design • Scatter plot with sweet-spot quadrant • Report writing	33.4%
Jordan Ling Shen Hang	• Data collection and Report writing • Map visualization with conditional formatting • EDA pbix with 3 analytical question pages • Data cleaning and data model relationships design	33.3%
Zainal Zaim Hakimi bin Zainal Effendy	• Data collection and Report writing • Bar chart for cost per GB of roaming data by brand • Travel persona classification design	33.3%

Declaration

We confirm that the contributions stated above accurately reflect the work carried out by each group member. We understand that false or misleading declarations may result in penalties and may be treated as academic misconduct.

Signatures

Student Name	Signature	Date
Sara Emilia Binti Sharifudin		23/5/2026
Jordan Ling Shen Hang		22/5/2026
Zainal Zaim Hakimi bin Zainal Effendy		24/5/2026